
FINAL EXAM QUESTIONS

1. Determine

$$\lim_{n \rightarrow \infty} \frac{\log n}{1 + n^2}$$

Solution : 0

2. Determine

$$\lim_{n \rightarrow \infty} \frac{\sin n}{1 + n}$$

Solution : 0

3. Determine

$$\lim_{x \rightarrow \infty} \frac{\sqrt[3]{x}}{\ln x}$$

Solution : ∞

4. Determine

$$\lim_{n \rightarrow \infty} \frac{(3n + 3)^2}{n(\sqrt{n} + 1)(\sqrt{n} - 1)}$$

Solution : 9

5. Determine

$$\lim_{n \rightarrow \infty} \frac{e^{2n}}{(1 + 2e^n)^2}$$

Solution : $\frac{1}{4}$

6. Determine

$$\lim_{t \rightarrow 0} \frac{e^{-1/t}}{\sqrt{t}}$$

Solution : 0

7. Calculate the derivative of the function

$$\frac{e^{-1/t}}{\sqrt{t}}$$

Solution :

$$e^{-1/t} \left(-\frac{1}{2t^{3/2}} + \frac{1}{t^{5/2}} \right)$$

8. Calculate the derivative of

$$\frac{-3 \sin(3x - 1)}{2\sqrt{\cos(3x - 1)}}$$

Solution :

$$\frac{-18 \cos(3x - 1) \sqrt{\cos(3x - 1)} - 9 \frac{\sin^2(3x - 1)}{\sqrt{\cos(3x - 1)}}}{4 \cos(3x - 1)}$$

9. Determine

$$\lim_{x \rightarrow 1} \frac{x^3 - 1}{x^2 - 1}$$

Solution : $3/2$

10. Calculate the derivative of

$$\sqrt{\ln \left(\frac{1}{x^2} \right) + 1}.$$

Solution : $-\left(\ln \left(\frac{1}{x^2} \right) + x \right)^{-1/2} \frac{1}{x}$

11. Calculate the derivative of the function

$$\sin(2e^{\sqrt{x}})$$

Solution : $\cos(2e^{\sqrt{x}}) e^{\sqrt{x}} \frac{1}{\sqrt{x}}$

12. Determine the first three terms of the Taylor expansion of order four of the function $(1 + 2x)^{-1/2}$ around $x = 0$.

Solution : $1 - x + \frac{3}{2}x^2$

13. Determine the first two terms of the second order Taylor expansion of the function $(1 + x)^{1/3}$ around $x = 0$.

Solution : $1 + \frac{x}{3}$

14. Determine the first two terms of the second order Taylor expansion of the function $\ln \left(\frac{1+x}{1-x} \right)$ around $x = 0$.

Solution : $2x$

15. Determine the first three terms of the Taylor expansion of order four of the function $\frac{x}{\sqrt{1-x^2}}$ around $x = 0$.

Solution : x

16. Calculate $\int_1^2 \frac{1}{1+2x} dx$.

Solution : $\frac{1}{2}(\ln(5) - \ln(3))$.

17. Calculate $\int_{-\pi/2}^{\pi/2} \cos(x) \sin(x) dx$.

Solution : 0.

18. Calculate the indefinite integral $\int e^x \cos(x) dx$.

Solution : $\frac{e^x}{2}(\sin(x) + \cos(x)) + C$

19. Calculate the indefinite integral $\int \frac{e^x}{(1+e^x)^2} dx$.

Solution : $-\frac{1}{1+e^x} + C$

20. Calculate $\int_0^1 x e^{x^2} dx$.

Solution : $\frac{e}{2} - \frac{1}{2}$

21. Calculate the indefinite integral $\int \frac{\ln x}{x} dx$.

Solution : $\frac{\ln^2 x}{2} + C$

22. Calculate the indefinite integral $\int \frac{1}{1+\sqrt{x}} dx$.

Solution : $2(1 + \sqrt{x} - \ln(1 + \sqrt{x})) + C$

23. Determine the dimension of the vector space spanned by the vectors

$$\begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix}.$$

Solution : 3

24. Find a basis of the subspace spanned by the following vectors

$$\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \\ -1 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \\ 0 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 5 \\ 5 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ -1 \\ 0 \\ 0 \end{pmatrix}$$

Solution :

$$\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \\ 0 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ -1 \\ 0 \\ 0 \end{pmatrix}$$

25. Consider the linear transformation in $\mathbb{R}^2$ which rotates every planar vector clockwise by 135° . Give the matrix of this transformation with respect to the standard basis.

Solution :

$$\begin{pmatrix} -1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}$$

26. Consider the plane in $\mathbb{R}^3$ determined by its basis vectors $(0, 1, -1)$ and $(0, 1/2, 1/2)$. Determine the orthogonal projection of the vector $(1, 1, 1)$ in the plane.

Solution : $(0, 1/\sqrt{2}, 1/\sqrt{2})$

27. Find a basis of the solution space of the system

$$\begin{cases} 3x_1 - x_2 + x_4 = 0 \\ x_1 + x_2 + x_3 + x_4 = 0. \end{cases}$$

Solution :

$$\begin{pmatrix} 1 \\ 3 \\ -4 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -2 \\ 1 \end{pmatrix}.$$

28. Compute the inverse of the following matrix $A = \begin{pmatrix} 1 & 3 & 0 \\ -1 & 2 & 4 \\ 0 & 5 & 1 \end{pmatrix}$.

Solution : $A = \begin{pmatrix} 6/5 & 1/5 & -4/5 \\ -1/15 & -1/15 & 4/15 \\ 1/3 & 1/3 & -1/3 \end{pmatrix}$

29. Compute the inverse of the following matrix $A = \begin{pmatrix} 0 & 1 & 2 \\ 2 & 4 & 3 \\ 2 & 3 & 6 \end{pmatrix}$.

Solution : $A^{-1} = \begin{pmatrix} -3/2 & 0 & 1/2 \\ 3/5 & 2/5 & -2/5 \\ 1/5 & -1/5 & 1/5 \end{pmatrix}$.

30. Compute the rank of the following matrix

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 3 & 6 & 3 \\ 1 & 1 & 2 & 4 \\ 2 & 3 & 5 & 0 \end{pmatrix}.$$

Solution : 3

31. Find the rank of the following matrix

$$\begin{pmatrix} 1 & -1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & 1 & -1 & -1 \end{pmatrix}$$

Solution : 3

32. Assume that the eigenvectors of a 2×2 matrix are $(1, 2)$ and $(0, 1)$ with corresponding eigenvalues 0 and 1. Determine the matrix.

Solution :

$$\begin{pmatrix} 0 & 0 \\ -2 & 1 \end{pmatrix}$$

33. Find the least square approximation of the function $f(x, y) = ax + by$ if three measurements give $f(0, 1) = 3$, $f(1, 2) = 8$ and $f(-1, 0) = 0$.

Solution : $\frac{x}{3} + \frac{11y}{3}$.

34. Find the closest line to the points $(0, 6)$, $(1, 0)$ and $(2, 0)$.

Solution : $y = -3x + 5$.

35. Find the least square approximation of the function $f(x) = ax^2 + b$ if three measurements give $f(0) = 0$, $f(1) = 1$ and $f(2) = 6$.

Solution : $a = \frac{20}{13}$, $b = \frac{-3}{13}$.

36. Find the eigenvalues and eigenvectors of the following matrix $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$.

Solution : $\lambda_1 = \frac{1+\sqrt{5}}{2}, \quad v_1 = (1, \frac{\sqrt{5}-1}{2}), \quad \lambda_2 = \frac{1-\sqrt{5}}{2} \quad v_2 = (1, -\frac{\sqrt{5}+1}{2})$

37. Determine the eigenvalues and eigenvectors of the matrix $A = \begin{pmatrix} 1 & 1 & 0 \\ 4 & 1 & 0 \\ 0 & 2 & 1 \end{pmatrix}$.

Solution : $\lambda_1 = 1, v_1 = (0, 0, 1), \lambda_2 = -1, v_2 = (-1/2, 1, -1), \lambda_3 = 3, v_3 = (1/2, 1, 1)$

38. Determine the symmetric matrix associated to the quadratic form in $\mathbb{R}^2$:

$$x^2 - xy - 3y^2.$$

Solution : $\begin{pmatrix} 1 & -1/2 \\ -1/2 & -3 \end{pmatrix}$

39. Consider a 2×2 matrix with eigenvectors $(1, -1)$ and $(0, 2)$ and corresponding eigenvalues 1 and 3 respectively. Determine the matrix.

Solution :

$$A = \begin{pmatrix} 1 & 0 \\ 2 & 3 \end{pmatrix}.$$

40. Determine whether the quadratic form $Q(x, y, z) = x^2 - 4xy + 4y^2 - 3z^2$ is positive definite, negative definite, positive semidefinite, negative semidefinite or indefinite.

Solution : indefinite

41. Determine whether the quadratic form in $Q(x, y, z) = xy + y^2 + 4xz + z^2$ is positive definite, negative definite, positive semidefinite, negative semidefinite or indefinite.

Solution : indefinite

42. Determine whether the matrix $\begin{pmatrix} -1 & 1 & 1 \\ 1 & -2 & 0 \\ 1 & 0 & -1 \end{pmatrix}$ is positive definite, positive semidefinite, negative definite, negative semidefinite, or indefinite.

Solution : indefinite

43. Determine whether the matrix $\begin{pmatrix} -1 & 2 & 0 \\ 1 & -4 & 0 \\ 0 & 0 & -3 \end{pmatrix}$ is positive definite, positive semidefinite, negative definite, negative semidefinite, or indefinite.

Solution : negative definite

44. Determine whether the matrix $\begin{pmatrix} 3 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$ is positive definite, positive semi-definite, negative definite, negative semidefinite, or indefinite.

Solution : positive definite

45. Find the minimum of the function $f(x) = x^x$ on $(0, \infty)$.

Solution : $\frac{1}{e}$

46. What is the norm of the gradient of $f(x, y) = (x - 2)^2 + (y - 1)^4$ in the point $(2, 1)$?

Solution : 0

47. Determine the determinant of the Hessian matrix of $f(x, y) = y^2e^x - xye^y$ at the point $(1, 1)$.

Solution : $-e^2$

48. Determine the critical points of the function $f(x, y) = x^3 + y^2 - 3x - 6y - 1$.

Solution : $(1, 3)$ and $(-1, 3)$

49. Determine if the critical points obtained in Question above are local maximum, local minimum or saddle points.

Solution : $(1, 3)$: local minimum $(-1, 3)$: saddle point

50. Determine if the following function is concave, convex or neither $f(x, y) = xy$.

Solution : neither

51. Find the critical points of the function $f(x, y) = e^{-(x^2+y^2)}e^{2x+4y}$

Solution : $(1, 2)$.

52. What is the norm of the gradient of $f(x, y) = (x - 2)^2 + (y - 1)^4$ at the point $(2, 1)$?

Solution : 0

53. Find all the stationary points of the function $f(x, y) = y^2e^{-x^2}$

Solution : $(x, 0)$

54. The variable cost of two products depends on the production outputs x and y as follows :

$$C(x, y) = 120x + \frac{1,200,000}{x} + 800y + \frac{32,000,000}{y}.$$

Determine the output which minimizes the cost function.

Solution : (100, 200)

55. Give the solution to the following differential equation

$$y'(t) = 0.1y(t), \quad y(0) = 2.$$

Solution : $y(t) = 2e^{0.1t}$

56. Give the solution to the following differential equation

$$y'(t) = e^t y(t), \quad y(0) = 2.$$

Solution : $y(t) = 2e^{e^t - 1}$